Midterm ~~2-1~~ 2016 - 2017

Pt:

1. a) The vector prod. is anticommutative. $(\vec{a}\times\vec{b} \neq \vec{b}\times\vec{a})$
(2017) The scalar prod. is symmetric. $(\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a} = \langle\vec{a},\vec{b}\rangle = \langle\vec{b},\vec{a}\rangle)$.

b) The triple scalar prod. $[\vec{a},\vec{b},\vec{c}]$ of $\vec{a},\vec{b},\vec{c}$ is equal to the directed volume of the parallelipiped constructed ~~to~~ on $\vec{a},\vec{b},\vec{c}$.

3. a) $\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\cdot\vec{b} - (\vec{a}\cdot\vec{b})\cdot\vec{c}$ (double vector prod.)
(2017)
b) Prove Jacobi:

$\rightarrow$ $\vec{a}\times(\vec{b}\times\vec{c}) + \vec{b}\times(\vec{c}\times\vec{a}) + \vec{c}\times(\vec{a}\times\vec{b}) = 0.$ (1)

$$\vec{a}\times(\vec{b}\times\vec{c}) = \cancel{(\vec{a}\cdot\vec{c})\cdot\vec{b}} - \cancel{(\vec{a}\cdot\vec{b})\cdot\vec{c}}$$

$$\vec{b}\times(\vec{c}\times\vec{a}) = \cancel{(\vec{b}\cdot\vec{a})\cdot\vec{c}} - \cancel{(\vec{b}\cdot\vec{c})\cdot\vec{a}}$$

$$\vec{c}\times(\vec{a}\times\vec{b}) = \cancel{(\vec{c}\cdot\vec{b})\cdot\vec{a}} - \cancel{(\vec{c}\cdot\vec{a})\cdot\vec{b}}$$

(+)

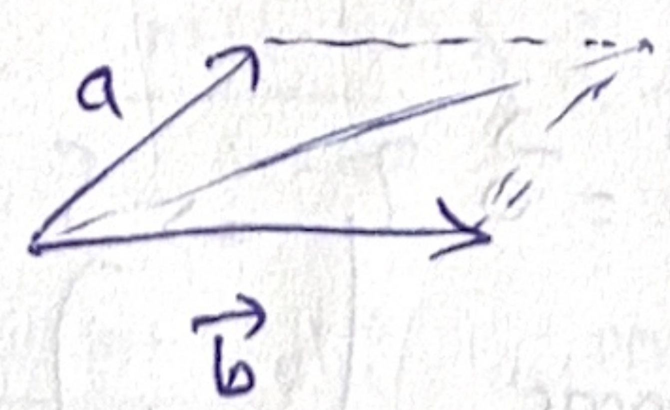
(1) = 0.

(scalar prod. is symm.)

1. a) $\Delta \rightarrow$ straight line and $A\in\Delta$
(2016)
2) $\vec{\Delta} = \{\vec{AH} \mid H\in\Delta\}$. $\rightarrow$ one-dimensional subspace of Δ.
$\hookrightarrow$ direction space of Δ.

b) norm of $\vec{a}\times\vec{b}$:

$$\|\vec{a}\times\vec{b}\| = \|\vec{a}\|\cdot\|\vec{b}\|\cdot\sin\measuredangle(\vec{a},\vec{b}). = A_{\text{parallelogram}}$$



2. b) Two straight lines are $\parallel \Leftrightarrow$ directions are orthogonal ($\perp$). F.
(2016)
Parallel lines must have parallel direction vectors, not orthogonal ones.

c) Two $\neq 0$ vectors are $\perp$ $\Leftrightarrow$ their dot prod. is zero. T.

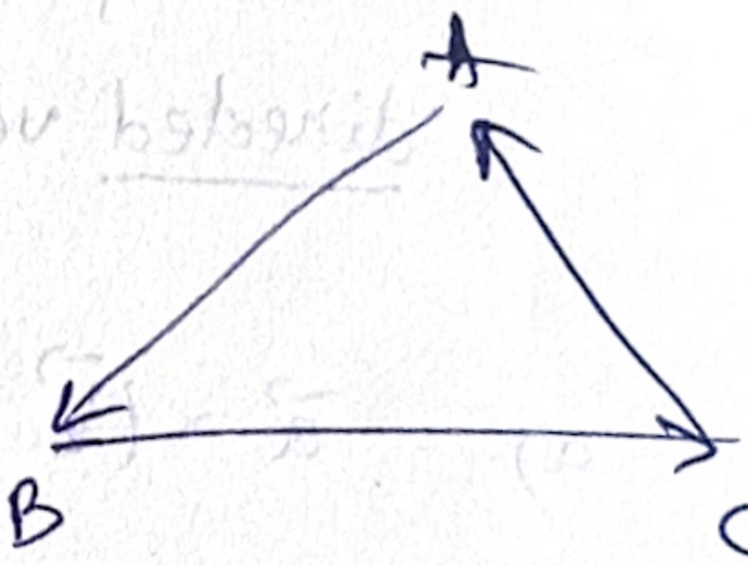
d) Scalar prod. is symm. T.

3. $\vec{a}, \vec{b}, \vec{c}$ noncollinear.

(2016) Necessary + sufficient condition for $\exists$ ΔABC ? $\boxed{\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a}}$

$\vec{BC} = \vec{a}, \quad \vec{CA} = \vec{b}, \quad \vec{AB} = \vec{c}.$

<u>Necessity</u>



If ΔABC $\exists$ $\Rightarrow$ $\vec{AB} + \vec{BC} + \vec{CA} = 0.$

$\Rightarrow \vec{a} + \vec{b} + \vec{c} = 0$

$\Rightarrow c = -(\vec{a} + \vec{b})$

$\vec{b}\times\vec{c} = \vec{b}\times(-\vec{a}-\vec{b}) = \underline{-\vec{b}\times\vec{a} - \vec{b}\times\vec{b} = \vec{a}\times\vec{b}}.$

$\vec{c}\times\vec{a} = -(\vec{a}+\vec{b})\times\vec{a} = -\vec{a}\times\vec{a} - \vec{b}\times\vec{a} = \vec{a}\times\vec{b}.$

$\Rightarrow \vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a}.$

<u>Sufficiency</u>. Assume $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a}.$

- $\Rightarrow \vec{a}\times\vec{b} - \vec{b}\times\vec{c} = \vec{0}$

  $\Rightarrow \vec{a}\times\vec{b} + \vec{c}\times\vec{b} = \vec{0}$

  $\Rightarrow \vec{b}\times(\vec{a}+\vec{c}) = \vec{0}. \Rightarrow (\vec{a}+\vec{c}) \parallel \vec{b} \Rightarrow \underline{\vec{a} + \vec{c} = \alpha\cdot\vec{b}}$ (2)

- $\vec{b}\times\vec{c} = \vec{c}\times\vec{a}$

  $\Rightarrow \vec{b}\times\vec{c} - \vec{c}\times\vec{a} = \vec{0}$

  $\Rightarrow \vec{b}\times\vec{c} + \vec{a}\times\vec{c} = \vec{0}$

  $\Rightarrow \vec{c}\times(\vec{b}+\vec{a}) = \vec{0} \Rightarrow \vec{c} \parallel (\vec{b}+\vec{a}) \Rightarrow \underline{\vec{a}+\vec{b} = \beta\cdot\vec{c}}$

  $\Rightarrow \vec{a} = \beta\vec{c} - \vec{b}.$ (1)

"(1) $\to$ (2)" $\Rightarrow \beta\vec{c} - \vec{b} + \vec{c} = \alpha\cdot\vec{b}$

$\left.\begin{array}{l} \Rightarrow (\beta+1)\vec{c} - (1+\alpha)\vec{b} = \vec{0} \\ \vec{b}, \vec{c} \text{ non-collinear} \end{array}\right\} \Rightarrow \begin{cases} \beta+1=0 & \Rightarrow \beta = -1 \\ \alpha+1=0 & \Rightarrow \alpha=-1 \end{cases}$

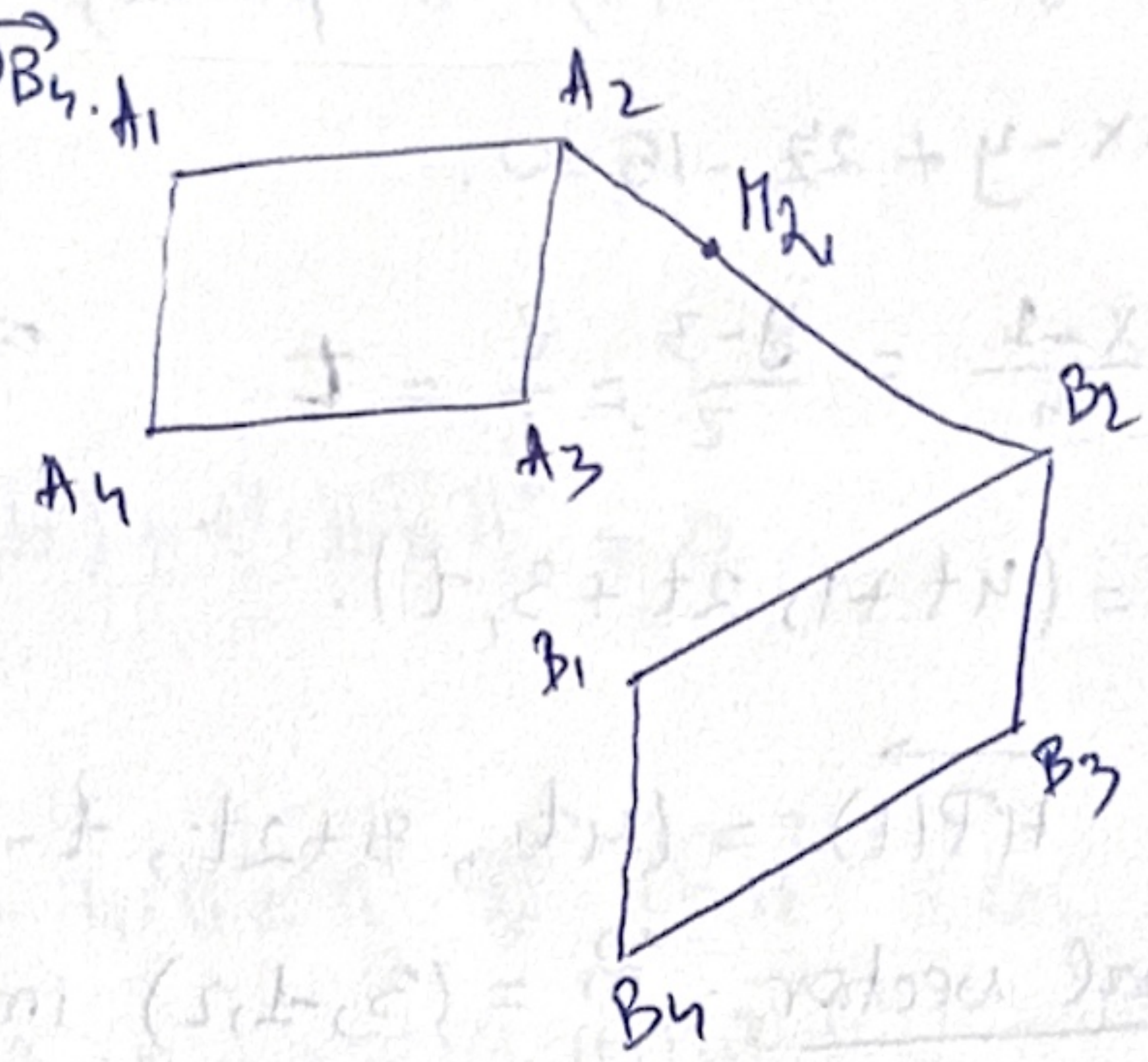
$\vec{a} + \vec{b} + \vec{c} = \vec{0}$

$\Rightarrow \boxed{\Delta \exists \Leftrightarrow \vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a}.}$

$A_1A_2A_3A_4$ par. $\Rightarrow \overrightarrow{OA_1}+\overrightarrow{OA_3} = \overrightarrow{OA_2}+\overrightarrow{OA_4}$

$B_1B_2B_3B_4$ par. $\Rightarrow \overrightarrow{OB_1}+\overrightarrow{OB_3} = \overrightarrow{OB_2}+\overrightarrow{OB_4}$.

$M_i \in A_iB_i$ s.t. $M_iB_i = \underset{\lambda}{3}A_iM_i$.



? $M_1M_2M_3M_4$ par

$$\boxed{\overrightarrow{OM_i} = \frac{\lambda}{\lambda+1}\overrightarrow{OA_i} + \frac{1}{\lambda+1}\overrightarrow{OB_i}}$$

$\lambda = 3$.

$\Rightarrow \overrightarrow{OM_i} = \frac{3}{4}\overrightarrow{OA_i} + \frac{1}{4}\overrightarrow{OB_i}$

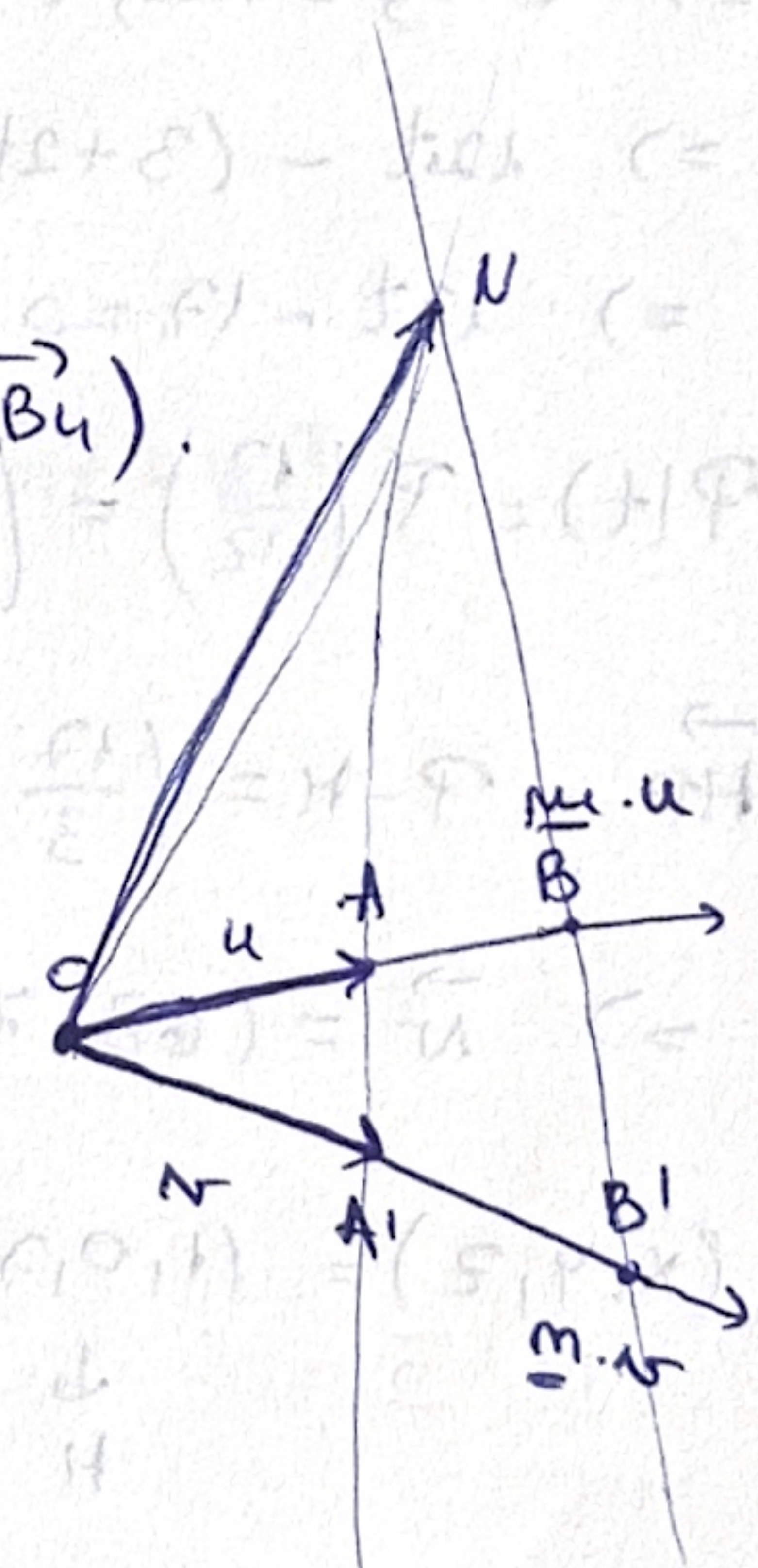
- $\underline{\overrightarrow{OM_1}+\overrightarrow{OM_3}} = \frac{3}{4}\overrightarrow{OA_1} + \frac{1}{4}\overrightarrow{OB_1} + \frac{3}{4}\overrightarrow{OA_3} + \frac{1}{4}\overrightarrow{OB_3}$

$$= \frac{3}{4}\underset{\parallel}{(\overrightarrow{OA_1}+\overrightarrow{OA_3})} + \frac{1}{4}\underset{\parallel}{(\overrightarrow{OB_1}+\overrightarrow{OB_3})}$$

- $\underline{\overrightarrow{OM_2}+\overrightarrow{OM_4}} = \dots = \frac{3}{4}(\overrightarrow{OA_2}+\overrightarrow{OA_4}) + \frac{1}{4}(\overrightarrow{OB_2}+\overrightarrow{OB_4})$.

$\Rightarrow M_1M_2M_3M_4$ par.

5) Intersection formula in an angle.



$$\boxed{\overrightarrow{ON} = \frac{m(m-1)}{m-m}\overrightarrow{OA} + \frac{m(m-1)}{m-m}\overrightarrow{OA'}}$$

$N \in AA'$: $\underline{\overrightarrow{ON} = (1-s)\cdot\vec{u} + s\cdot\vec{v}}$. (1)

$N \in BB'$: $\overrightarrow{ON} = m\cdot\vec{u} + t\cdot(m\vec{v} - m\vec{u})$

$\underline{\overrightarrow{ON} = m(1-t)\vec{u} + m.t.\vec{v}}$ (2)

$(1) = (2) \Rightarrow \begin{cases} (1-s) = m(1-t) \\ s = m.t \end{cases} \Rightarrow 1 - mt = m(1-t)$

$\Rightarrow 1 - m = -mt + mt$

$\Rightarrow 1 - m = (m-m)t$

$\Rightarrow t = \frac{1-m}{m-m}$.

$s = m.\frac{1-m}{m-m}$.

(1): $\overrightarrow{ON} = (1-s)\cdot\vec{u} + s\cdot\vec{v}$

$$= \left(1 - \frac{m(1-m)}{m-m}\right)\vec{u} + \frac{m(1-m)}{m-m}\vec{v}$$

$= \dots$

$$= \frac{m(m-1)}{m-m}\underset{\parallel}{\vec{u}} + \frac{m(m-1)}{m-m}\underset{\parallel}{\vec{v}}$$

$= \dots \cdot \overrightarrow{OA} + \dots \cdot \overrightarrow{OA'}$

6) Line through $M(1,0,7)$ $\parallel$ plane, $\cap$ line.

$\pi: \ 3x - y + 2z - 15 = 0.$

$d: \ \frac{x-1}{4} = \frac{y-3}{2} = \frac{z}{1} = t \quad \Rightarrow x = 4t+1$

$\Rightarrow y = 2t+3$

$\Rightarrow P(t) = (4t+1, 2t+3, t). \quad \Rightarrow z = t$

? dir. vector: $\overrightarrow{MP(t)} = (4t, 3+2t, t-7)$ must be orthogonal to the normal vector $\vec{n} = (3,-1,2)$ in order for ~~d to~~ the line to be $\parallel \pi$.

$\Rightarrow (4t, 3+2t, t-7)\cdot(3,-1,2) = 0$

$\Rightarrow 12t - (3+2t) + 2(t-7) = 0$

$\Rightarrow 12t - 17 = 0 \Rightarrow t = \frac{17}{12}$

$P(t) = P\left(\frac{17}{12}\right) = \left(1 + \cancel{4}\cdot\frac{17}{\cancel{12}_3}, \cancel{2}\cdot\frac{17}{\cancel{12}_6} + 3, \frac{17}{12}\right) = \left(\frac{20}{3}, \frac{35}{6}, \frac{17}{12}\right)$

$\overrightarrow{MP} = P - M = \left(\frac{17}{3}, \frac{35}{6}, -\frac{67}{12}\right) \ \Big|\cdot 12$

$\Rightarrow \vec{v} = (68, 70, -67).$

$l: (x,y,z) = \underbrace{(1,0,7)}_{M} + \lambda\cdot(68,70,-67).$

$\Rightarrow$

$$\frac{x-1}{68} = \frac{y-0}{70} = \frac{z-7}{-67}$$

7) Locus points equidistant from plane xOy and $C(0,0,1)$.

$d(P, xOy) = |z|.$

$d(P,C) = \sqrt{x^2+y^2+(z-1)^2} = \sqrt{(x-x_C)^2+(y-y_C)^2+(z-z_C)^2} = |z| \ \Big|^2$

$\Leftrightarrow x^2+y^2+(z-1)^2 = z^2$

$\Rightarrow x^2+y^2+\cancel{z^2}-2z+1 = \cancel{z^2}$

$\Rightarrow x^2+y^2-2z+1 = 0$

$\Rightarrow z = \frac{x^2+y^2+1}{2}$

a) Prove Cauchy-Schwartz.

$$|\vec{a}\cdot\vec{b}| \le \|\vec{a}\|\cdot\|\vec{b}\|.$$

$$f(t) = \|\vec{a} - t\cdot\vec{b}\|^2 \geq 0$$

$$= \|\vec{a}\|^2 - 2t\cdot(\vec{a},\vec{b}) + t^2\cdot\|\vec{b}\|^2 \geq 0$$

$$\Rightarrow \Delta \le 0$$

$$\Delta = (-2(\vec{a},\vec{b}))^2 - 4\cdot\|\vec{a}\|^2\cdot\|\vec{b}\|^2 \le 0$$

$$\Rightarrow 4(\vec{a},\vec{b})^2 \le 4\|\vec{a}\|^2\cdot\|\vec{b}\|^2 \;|:4$$

$$\Rightarrow (\vec{a},\vec{b})^2 \le \|\vec{a}\|^2\cdot\|\vec{b}\|^2$$

$$\Rightarrow |\vec{a}\cdot\vec{b}| \le \|\vec{a}\|\cdot\|\vec{b}\|$$

b) Deduce ineq:

$$(a_1b_1 + a_2b_2 + a_3b_3)^2 \le (a_1^2 + a_2^2 + a_3^2)\cdot(b_1^2 + b_2^2 + b_3^2)$$

$$\vec{a} = (a_1, a_2, a_3);\; \vec{b} = (b_1, b_2, b_3)$$

$$\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

$$\|\vec{a}\|^2 = a_1^2 + a_2^2 + a_3^2$$

$$\|\vec{b}\|^2 = b_1^2 + b_2^2 + b_3^2$$

$$\Rightarrow (\vec{a}\cdot\vec{b})^2 \le \|\vec{a}\|^2\cdot\|\vec{b}\|^2$$